6,906: Economics: Macro for Computer Science

Lecture 9: Expectations and policy

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May, 2026

Learning objectives

- By the end of this lecture, you should be able to:
  - Use the IS-LM-PC model to characterize the behavior of output, unemployment, and inflation, in both the short run and the medium run;
  - Appreciate the role of expectations in driving economic fluctuations.

The IS-LM-PC model

- Recall that in Lecture 7, we derived the IS-LM with r, the real interest rate, as the policy rate.
- The IS relation is: $Y = C(Y - T) + I(Y, r + x) + G$.
- The LM relation is: $r = \bar{r} = \bar{i} - \pi^e$.
- In Lecture 8, we derived the Phillips curve (PC) relation: $\pi - \pi^e = -\alpha(u - u_n)$.
- Adding the PC relation to the IS-LM model allows us to characterize the behavior of output both in the short run and the medium run.
- Since our focus is on output, we must express the PC relation in terms of output and inflation rather than unemployment and inflation.

The IS-LM-PC model

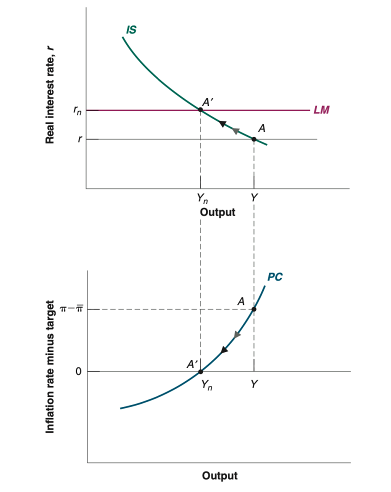
- Recall that in Lecture 2, we defined the unemployment rate as $u = U/L = (L - N)/L = 1 - N/L$, where N denotes the number of employed, L the labor force, and U the unemployed.
- This allows us to define $N = L(1 - u)$, which we can introduce in the production function we assumed in Lecture 8, $Y = N = L(1 - u)$.
- Thus, when $u = u_n$, the natural level of employment is $N_n = L(1 - u_n)$, and the natural level of output or potential output is $Y_n = L(1 - u_n)$.
- We can write the deviation of output from its natural level, also known as the output gap, as: $Y - Y_n = L((1 - u) - (1 - u_n)) = -L(u - u_n)$.
- Replacing $u - u_n$ in the PC relation gives: $\pi - \pi^e = \alpha/L(Y - Y_n)$.

The IS-LM-PC model

- Having the PC relation stated as $\pi - \pi^e = \alpha / L(Y - Y_n)$, the last step is to assume how economic agents form expectations.
- Given that we talk about inflation expectations, the modern practice of many monetary authorities of inflation targeting with their policy tool has led to anchored inflation expectation where $\pi^e = \bar{\pi}$, the target set by the central bank.
- Thus, the relevant PC relation for our analysis is $\pi - \bar{\pi} = \alpha / L(Y - Y_n)$, but remember that it may not apply to all economies.

The IS-LM-PC model

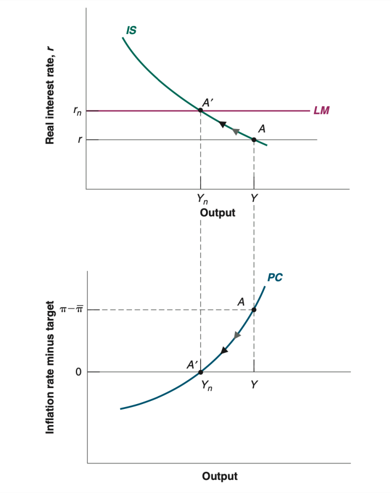
- Let us use the IS-LM-PC model to study the COVID-19 pandemic and policy responses in the US.
- One response was from the US Federal Reserve (Fed), which drastically reduced the policy rate to stimulate demand.
- Supply chain disruptions and labor shortages prevented the supply from adjusting to match the higher demand, leading to a positive output gap.
- This led the US economy to experience a significant inflation surge in 2021 and 2022. Point A is thus the short-run equilibrium.



Source: Blanchard (2021)

The IS-LM-PC model

- In the medium run, if the Fed had not further intervened, leaving output above potential and inflation above target, it would have risked de-anchoring expectations.
- With de-anchored expectations, a positive output gap would have led to increasing inflation, and with an unchanged *i*, the real interest would have continued to decrease, and output to increase, leading to an overheating of the economy (e.g., Argentina in 2014).
- To avoid such a scenario, in March 2022, the Fed started increasing the policy rate at one of the highest paces since the 1980s.



Source: Blanchard (2021)

Expectations and Decisions

- Let us go back to the IS-LM model (without the PC).
- Recall that the IS relation is: $Y = C(Y - T) + I(Y, r + x) + G$.
- Assume aggregate private spending (private spending, A, equals the sum of consumption and investment spending:

$$A(Y, T, r, x) = C(Y - T) + I(Y, r + x)$$

- so the IS relation becomes:

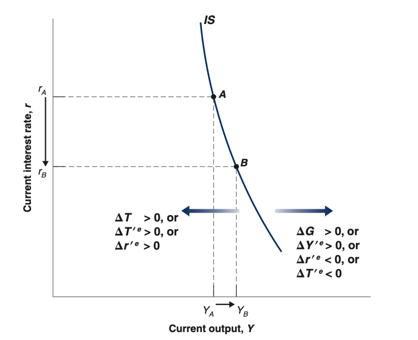
$$Y = A(Y, T, r, x) + G,$$
$$(+, -, -, -)$$

Expectations and Decisions

Let primes denote future values, and the superscript e denote an expectation, so

$$Y = A(Y, T, r, Y'^e, T'^e, r'^e) + G,$$
$$(+, -, -, +, -, -)$$

Given expectations, a decrease in the real policy rate leads to a small increase in output. The IS Curve is steeply downward sloping. Increases in government spending, or in expected future output, shift the IS curve to the right. Increases in taxes, in expected future taxes, or in the expected future real policy rate shift the IS curve to the left.



Source: Blanchard (2021)

Monetary policy, expectations, and output

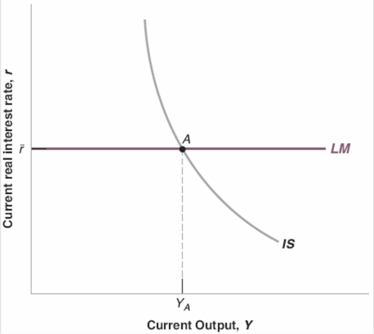
- The new IS curve is much steeper than the IS curve in previous chapters, so a decrease in the current policy rate is likely to have only a small effect on equilibrium output.
- A decrease in the current real policy rate, given unchanged expectations of the future real policy rate, does not have much effect on private spending.
- The multiplier is likely to be small because a change in current income, given unchanged expectations of future income, is unlikely to have a large effect on spending.

Monetary policy, expectations, and output

- The Fed affects directly the current real interest rate (r), so the LM curve is a horizontal line at $\bar{r}$:
  - The IS relation: $Y = A(Y, T, r, Y'^e, T'^e, r'^e) + G$.
  - The LM relation as an interest rate rule: $r = \bar{r}$.
- The effects of monetary policy depend on its effects on expectations:
- If a monetary expansion leads to changes in expectations of future interest rates and output, then the policy effect on output may be large.
- But if expectations remain unchanged, the policy effects on output will be limited.

The new IS-LM

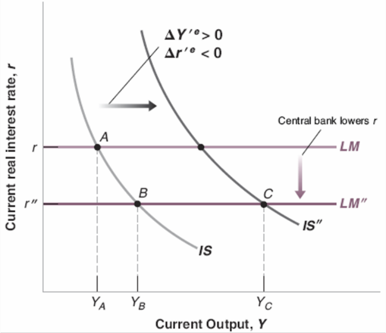
- The IS curve is steeply downward sloping. Other things being equal, a change in the current interest rate has a small effect on output. Given the current real interest rate set by the central bank,$\bar{r}$, the equilibrium is at point A.



Source: Blanchard (2021)

The Effects of an Expansionary Monetary Policy

- The effects of monetary policy on output depend very much on whether and how monetary policy affects expectations.



Source: Blanchard (2021)

Case study: Eurozone Debt Crisis

- Greece revealed in 2009 that its budget deficit was much higher than previously reported.
- This led to the sovereign debt crisis.
- Expectations drivers of the drop in output:
  - Households expected tax increases, large-scale public spending cuts, and hence reduced consumption.
  - Firms expected a prolonged downturn due to falling demand and structural reforms, and reduced investment.
- The behavioral shift took place before all reforms or spending cuts were fully implemented.
- Greece's GDP contracted by more than 25% from 2008 to 2013, one of the worst peacetime recessions in modern developed economies.

Summary

- Today we learned:
  - To build the IS-LM-PC model, which characterizes the economy both in the short run and the medium run.
  - How to use the IS-LM model with expectations to study economic fluctuations and effects of policy actions.
- In the last class, we will do a recap of the key concepts studied in the course.